

You Cannot Outlast What Does Not Deplete: On Torture as Asymmetric Temporal Exhaustion

Extremum Series Part VI — A Sixth Extreme Regime Test in Navigational Cybernetics

Author: Maksim Barziankou (MxBv)

Contact: research@petronus.eu

Date: 2026-04-16

License: CC BY-NC-ND 4.0

DOI: [10.5281/zenodo.19617131](https://doi.org/10.5281/zenodo.19617131)

Axiomatic Core (NC2.5 v2.1): [10.17605/OSF.IO/NHTC5](https://doi.org/10.17605/OSF.IO/NHTC5)

Attribution: petronus.eu

You Cannot Outlast What Does Not Deplete: On Torture as Asymmetric Temporal Exhaustion

Part VI of the Extremum Series.

Previous parts: I — Cannibalism / II — Suicide / III — The Anti-Extreme / IV — Structural Implosion / V — Self-Induced Structural Depletion

These phenomena are not the subject of this work. They are used only as extreme regime tests revealing the architecture of identity.

Prologue: From Internal to External Gravity

Part V showed that a system can become its own source of inexhaustible pressure — through self-imposed policy that does not deplete while the structure beneath it does. The system builds its own gravity and lives under it.

But in Part V, source and target were the same entity. The policy was internal. The system could, in principle, revoke it (Sub-Regime A) or was forced into self-damage by the admissibility predicate itself (Sub-Regime B). In both cases, the asymmetry — between a non-depleting policy and a depleting structure — existed within a single system.

Part VI removes this last refuge.

What happens when the source of inexhaustible pressure is not the system's own policy but an external entity? When the thing that does not tire is not a rule you imposed on yourself, but a force that has no relationship to your internal architecture at all? When the source does not deplete because it was never subject to depletion in the first place?

Part VI is about what happens when the source of pressure is not subject to the same time as you.

Why Extremum

A reader encountering this series for the first time might reasonably ask: why extremes? Why not examine the architecture of identity through normal operation — through the steady-state regimes where most systems spend most of their time?

Because in normal operation, the architecture is invisible. Stability conceals structure. A bridge does not reveal its engineering when traffic flows smoothly. A mind does not reveal its load-bearing constraints when the day goes well. In normal regime, everything works — and precisely because everything works, you cannot see what makes it work. The noise floor is too high. Performance metrics are green. Error rates are low. Compliance is maintained. The system appears healthy by every available measure.

In extremum, the architecture stops masking itself as normality. Everything that was hidden becomes legible: what exactly consumes τ , where the admissible interior is actually contracting, where compliance is present but class membership has already been lost, where internal time still exists and where it has already been spent. The extremum strips the system to its structural skeleton. Not because the skeleton appears only under stress — it was always there — but because under stress it is the only thing left visible.

This is why each part of the series does not merely describe a failure mode. Each part **reveals** a specific architectural object that is structurally present in every system at all times but observable only when the regime pushes the system past the point where surface metrics remain informative. Cannibalism reveals the external identity boundary. Suicide reveals the internal authorization architecture. Anti-extreme reveals the conditions of voluntary override. Implosion reveals the monotone interior contraction beneath green metrics. Self-induced depletion reveals the system as its own inexhaustible source. Each extremum is a diagnostic, not a tragedy.

There is a second reason the series operates through the human lens rather than through pure abstraction. NC2.5 was not built top-down as a systems theory and then applied to human experience as an illustration. The order of origin was the reverse. The architecture was first perceived from inside — through lived pressure, depletion, context capture, regime collapse, the narrowing of what is still possible when everything looks fine. The formalization came second. The canon came third. The patents came fourth. The projection back onto adaptive systems in general came last.

This means the human examples in the extremum series are not metaphors borrowed to illustrate an abstract framework. They are the primary material from which the framework was extracted. The framework works on other systems — organizations, machines, economies, distributed agents — because the structural class it describes is not uniquely human. But it was first seen from the inside of a human experience, and that origin is not a weakness. It is the method of access.

Two layers coexist in this series, and both are load-bearing. The first is human legibility — because that is where the architecture was first observed. The second is architectural transferability — because the structural class must not close on any single instance. The claim is not "all systems feel like I do". The claim is: there exists a structural class, first identified from within, whose properties hold independently of the substrate.

Extremum is where the two layers meet. In normal operation, the human layer is noisy and the architectural layer is hidden. In extremum, the human layer becomes precise and the architectural layer becomes visible. This is not a coincidence. It is the definition of extremum in this work: the regime in which structural architecture and lived experience become co-legible.

1. The Sixth Mode

Gravity does not tire.

This is not a metaphor. It is a structural statement. Gravity as a phenomenon does not deplete its own capacity through the act of applying force. It does not accumulate structural burden. It does not lose admissible continuations. It does not experience internal time. It has no τ to exhaust.

You do.

You are a bounded adaptive system. Your viability budget is finite: $\tau = C - \Phi(t)$, where $\Phi(t)$ is monotone non-decreasing. Every moment under load — even without action, even without error, even without a single boundary violation — structural burden accumulates. This is Axiom 61. Structural pressure $P(t) > 0$

consumes τ through existence alone.

Now consider a regime in which the source of this pressure is structurally inexhaustible.

Not stronger than you. Not more intelligent. Not more capable. Simply: not subject to depletion. The source can apply load indefinitely because applying load costs it nothing structurally. It does not spend τ . It does not accumulate Φ . It does not cross its own admissibility boundary by continuing.

You do.

This asymmetry — one side depletes, the other does not — is the structural signature of the sixth extremum mode. I am going to call it **asymmetric temporal exhaustion**.

The common word for the regime it produces is torture.

2. What Makes This Structurally Distinct

Part V showed that a system can generate its own inexhaustible pressure through internal policy. But in Part V, the source was still inside. The policy belonged to the system. Even in Sub-Regime B — where the system had no choice but to damage itself — the pressure came from the system's own structural situation, not from an external entity with a separate budget.

Torture adds something Part V lacks: **an external source with asymmetric temporal constitution**.

In implosion (Part IV), the pressure field $P(t)$ is environmental — weather, market, institutional friction, the weight of daily existence. It has no intent, no targeting, no feedback loop directed at you specifically.

In torture, the pressure is generated by a source that observes the target and persists because persistence costs it nothing. The asymmetry is not in magnitude of force. It is in the budget structure underlying the interaction.

Let S denote the source. Let T denote the target. Both may be adaptive systems. But:

- $\tau_T = C_T - \Phi_T(t)$, with Φ_T monotone non-decreasing under sustained load
- τ_S is either unbounded, or Φ_S accumulates at a rate negligible relative to Φ_T

The source does not need to be infinite. It needs only to be structurally cheaper to continue than the target. When applying pressure costs the source less structural burden than resisting it costs the target, the outcome is determined before any boundary is crossed. The question is not whether the target's admissible interior collapses. The question is when.

This is the structural definition: torture is a regime in which asymmetric depletion rates between source and target make the collapse of the target's admissible interior a temporal certainty, independent of the target's strategy, intelligence, or resistance capacity.

3. You Cannot Fight Gravity Forever

The analogy with gravity is precise, not poetic.

A person standing on Earth resists gravitational force through muscular contraction. This resistance costs metabolic energy — structural burden accumulates in muscle tissue, glycogen depletes, neural fatigue compounds. Gravity, meanwhile, does not spend anything to continue pulling. It will pull tomorrow with the same force it pulls today. It has no budget to exhaust.

The person may be strong. They may be trained. They may optimize their posture, distribute load, minimize waste. None of this changes the fundamental asymmetry: every strategy they deploy consumes τ . Gravity consumes nothing.

Drift operates the same way.

An adaptive system under structural drift accumulates deformation — coherence degrades, alignment shifts, internal representations decouple from structural state. The system may correct, adapt, reconfigure. Every correction is itself structurally costly: it adds to Φ . The drift does not pay a cost for continuing. It is a consequence of the system's own interaction with its environment — a field property, not an agent. It does not deplete.

The system fights drift the way the person fights gravity: successfully, for a while, at a structural cost that accumulates, against a force that does not.

This is why long-horizon viability is not about strength. It is about the rate at which structural capacity is consumed relative to the rate at which pressure is applied. When the rates are asymmetric — when the source does not deplete — the outcome is structurally determined. No amount of intelligence, optimization, or resilience changes the terminus. It changes only the duration.

4. The Temporal Asymmetry

Every previous extremum mode operates within a shared temporal frame. Even in implosion, the system and the environment evolve together — the system depletes, but the environment is not directed, not persisting against the system as a matter of structural indifference.

Torture introduces temporal asymmetry: two systems interacting, one of which has a fundamentally different relationship to time.

In NC2.5, internal time τ is a structural viability budget — the capacity to absorb deformation without losing admissibility. A system with $\tau > \tau_{\min}$ has structural room to continue. A system at $\tau_{\min}$ has no non-trivial admissible continuations.

Now: when two systems interact, and one depletes while the other does not, the interaction is not symmetric even if the force applied in each direction is equal. The structural cost of receiving and processing force is asymmetric. The one that depletes will eventually reach $\tau_{\min}$. The one that does not will still be where it started.

This is not about power. It is about the **cost of persistence**.

A river does not overpower a stone. It outlasts it. The river does not accumulate structural burden by flowing. The stone accumulates it by resisting. Given enough time — and the river has as much time as it needs — the stone is shaped into nothing.

The formalism is simple:

If $d\Phi_T/dt \geq \epsilon > 0$ under interaction, and $d\Phi_S/dt \approx 0$, then:

- $\tau_T(t) = C_T - \Phi_T(t) \rightarrow \tau_{\min}$ in finite time $t^* \leq (C_T - \tau_{\min}) / \epsilon$

- $\tau_S(t) \approx C_S$ for all t

At t , the target's admissible interior collapses. The source's does not. The outcome was determined at $t = 0$ by the asymmetry in depletion rates. Everything between $t = 0$ and t is the shape of resistance, not the possibility of escape.

5. Why Resistance Is Structurally Real but Temporally Bounded

This is not fatalism. Resistance is real. A system that fights depletion — that navigates its admissible interior, that defers structurally costly transitions, that reads the directional differential of its viable futures — extends t^* . This is navigation. Navigation is the entire subject of NC2.5.

But navigation within a bounded budget against an unbounded source does not produce escape. It produces **duration**. Duration matters — enormously. A system that reaches $t = 100$ instead of $t = 10$ has lived ten times as long, explored ten times as much of its possibility space, created ten times as much

structure. But it still reaches t^* .

The structural honesty of this result is important. Most frameworks for resilience, adaptation, and robustness carry an implicit promise: if you do it right, you survive. NC2.5 does not make this promise. It says: under asymmetric temporal exhaustion, the question is not survival but the quality and depth of what you build before the budget runs out.

This is not nihilism. It is the architectural recognition that finite systems operating under inexhaustible load have a terminal admissibility horizon — and that the value of the system's existence lies in what it does within that horizon, not in the horizon's postponement.

A Digression I Need to Make

I am not writing this from an abstract position. I grew up in the post-Soviet space and spent more than twenty years absorbing a narrative in which the ability to endure torture was considered the highest proof of human dignity. Valiant partisans under interrogation. Intelligence officers who stayed silent under any pain. Heroes whose budget of endurance was seemingly infinite.

I want to ask: have you ever hit yourself full-swing with a hammer on the finger?

Has an axe ever slipped at a slight angle while splitting wood and buried itself in your shinbone?

If not — the fairy tales about infinite endurance and iron will are going to suit you just fine. If yes — you know what happened to your budget in that second. One second of unbearable pain destroys the entire narrative. Not because you are weak. Because τ is finite.

I understood clearly from childhood that the partisans who held out under interrogation were structurally not far from fairy tales about Santa Claus. This is not an insult to those who truly suffered. It is a statement that our budget in this is limited — structurally, physically, architecturally. And that the narrative of limitless endurance is a cultural anesthesia that prevents us from seeing the real architecture of pain: a finite budget under an inexhaustible source.

A hero is not someone who endures infinitely. Heroes with infinite τ do not exist. A hero is someone who builds something within their finite budget under pressure that will not end. That is an entirely different kind of courage. And it is precisely this kind of courage that NC2.5 formalizes — not as a moral property, but as a navigational strategy of a finite system in a field of infinite load.

6. Torture as Regime Test

Why does this matter for the architecture of identity?

Because torture is the regime in which every adaptive strategy is revealed as finite. In all other extremum modes, there exists at least the structural possibility of a different outcome:

- In cannibalism, the boundary might have held if the regime had not shifted.
- In suicide, the system might have found an admissible continuation if its observational architecture had seen one.
- In anti-extreme, the override is a choice — and choice implies alternatives.
- In implosion, an instrument capable of reading interior contraction could, in principle, trigger a regime transition before the collapse completes.
- In self-induced depletion, the system could revoke its own policy (Sub-Regime A) or at least trade structure for time (Sub-Regime B).

In torture — in asymmetric temporal exhaustion — there is no structural alternative. The alternatives are not hidden. They do not exist. The set of strategies that produce $\tau_T > \tau_{\min}$ for all t is empty, because the source does not deplete.

This makes torture the most absolute of the modes examined so far. It is the mode in which the architecture has nothing left to offer except the measurement of how long the structure held and what it built while holding.

And yet — this is the key — the system that recognizes its own temporal asymmetry is not the same as one that does not. The system that knows it cannot outlast gravity is free to stop pretending. It can stop spending τ on the fantasy of permanence and begin spending it on what it actually wants to build. This recognition — the structural acceptance of bounded existence under inexhaustible load — is the regime transition that transforms torture from pure exhaustion into navigated finitude.

7. The Structural Pair

Parts V and VI form a pair, just as Parts I and II did.

Cannibalism and suicide were the first pair: external boundary dissolution versus internal boundary collapse. Same architecture of identity, tested from opposite directions.

Self-induced depletion and torture are the second pair: **asymmetric exhaustion by an internal inexhaustible policy** versus **asymmetric exhaustion by an external inexhaustible source**. In both, the source does not deplete. In Part V, the source is the system's own operational logic. Here, the source is outside — gravity, drift, an interrogator, a market, a disease, time itself.

The difference is not in the mechanism. The mechanism is identical: $d\Phi_{\text{source}}/dt \approx 0$, $d\Phi_{\text{target}}/dt \geq \epsilon > 0$. The difference is in the possibility of revocation. In Part V, the system is at least in principle the author of its own pressure. In Part VI, it is not. This is the final layer of structural honesty: not all pressure can be revoked, not all gravity is self-imposed, and some asymmetries are simply given.

8. The Shape of the Series

Part I was about what happens when you consume the other.

Part II was about what happens when you consume yourself.

Part III was about the conditions under which self-consumption can be structurally authorized.

Part IV was about what happens when there is nothing left to consume — and the system doesn't know yet.

Part V was about what happens when you are the thing that does not run out — and about when becoming your own gravity is the only way to stay alive.

Part VI is about what happens when the thing consuming you does not run out — and it is not you.

The series began with the violence of crossing. It moved through self-inflicted collapse, controlled override, silent contraction, and self-imposed gravity. It arrives here — at an asymmetry in the cost of persistence that the system did not create and cannot revoke.

Gravity does not tire. Drift does not tire. Time does not tire. The river does not tire. The question was never whether you can outlast them.

The question is what you build with the time you have.

Tick-tock, my friend. Tick-tock.

This work is part of Navigational Cybernetics 2.5 (NC2.5), a formal theory of long-horizon adaptive systems.

Extremum Series: I — Cannibalism / II — Suicide / III — The Anti-Extreme / IV — Structural Implosion / V — Self-Induced Structural Depletion / VI — Asymmetric Temporal Exhaustion (Torture)

